To what extent does pH variation influence the antimicrobial efficacy of natural preservatives (e.g., neem extract, rosemary oil) compared to synthetic preservatives (e.g., sodium benzoate, parabens)?

Research Paper By Yashasavini Arora

Abstract

Preservatives are added to food, cosmetics, and medicines to stop the growth of harmful microorganisms and keep products safe and long-lasting. This study explores how pH variation affects the antimicrobial effectiveness of both natural preservatives (neem extract and rosemary oil) and synthetic preservatives (sodium benzoate and parabens). Using lab experiments on topical cream formulations, the antimicrobial activity was measured across pH levels 4.0, 5.5, 6.5, and 7.5.The results showed that all four preservatives worked best in acidic conditions (especially pH 4.0 to 5.5) and became less effective as pH increased. Sodium benzoate had the highest antimicrobial activity at low pH but lost strength in alkaline conditions. Neem extract and rosemary oil also showed good results in acidic pH, though slightly weaker than synthetics. Parabens were more consistent across a broader pH range.

This study highlights the importance of choosing preservatives based on the product's pH to ensure safety and stability. It also supports growing interest in natural alternatives and calls for more real-world testing to better understand their performance.

Keywords: pH variation, natural preservatives, neem extract, rosemary oil, sodium benzoate, parabens, antimicrobial efficacy, cosmetic formulations, product safety, topical creams.

1. Introduction

1.1 Background

Preserving food, cosmetics, and pharmaceutical products is an essential part of modern life. Without proper preservation, these products can spoil quickly due to microbial growth, making them unsafe to use or consume. Microorganisms such as bacteria, yeasts, and molds are naturally present in our environment. When they grow inside food or topical products like creams or lotions, they can cause infections, bad smell, colour changes, or complete product damage. This not only leads to wastage but can also be harmful to human health.

In the food industry, spoilage not only affects taste and appearance but also puts consumers at risk of food poisoning. Similarly, in cosmetics and pharmaceutical products, microbial contamination can lead to skin infections, irritation, or loss of product effectiveness. Therefore, preventing microbial spoilage is a top priority for product manufacturers to ensure safety, increase shelf life, and maintain product quality.

Over the years, different techniques have been used to reduce or prevent spoilage[1]. These include physical methods such as refrigeration, drying, and heat treatment, as well as chemical methods like the addition of preservatives. Among these, using preservatives is one of the most effective ways to stop or slow down the growth of unwanted microorganisms[2]. This helps in

---

[1] Tang, Z., & Du, Q. (2024). Mechanism of action of preservatives in cosmetics. *Journal of Drug Science and Cosmetic Technology*, *1*, Article 100054. https://doi.org/10.1016/j.jdsct.2024.100054

[2] Shelke, O. S. (2018). *Effect of pH on the efficacy of the sodium benzoate as antimicrobial preservative in topical formulation*. Sinomune Pharmaceutical Co. https://www.researchgate.net/publication/382696275

maintaining the safety and quality of products throughout their expected shelf life, even after opening.With the increasing demand for safe and long-lasting products, preservatives play a vital role in ensuring consumer protection, especially in products that cannot be kept under refrigeration all the time, such as cosmetic creams, lotions, or ready-to-eat foods.

1.2 Role of Preservatives

Preservatives are substances added to products to prevent the growth of harmful microorganisms. They help extend shelf life and reduce the risk of infections or spoilage. In both the food industry and cosmetic or pharmaceutical sectors, preservatives are widely used to keep products safe and effective during storage and use.There are two main types of preservatives: natural and synthetic. Natural preservatives are obtained from plants, animals, or microbes. Examples include neem extract and rosemary oil, both known for their antimicrobial and antioxidant properties[3]. They are becoming popular because of growing consumer demand for "natural" and "chemical-free" products.

On the other hand, **synthetic preservatives** are man-made chemical compounds designed to prevent microbial growth. Sodium benzoate and parabens are common examples. These preservatives are effective, affordable, and widely used, especially in products where natural alternatives may not perform well[4]. While synthetic preservatives have been effective for

---

[3] ScienceDirect. (2024). *Natural preservatives*. In *Pharmacology, toxicology and pharmaceutical science*. https://www.sciencedirect.com/topics/pharmacology-toxicology-and-pharmaceutical-science/natural-preservatives

[4] GERDHelp. (n.d.). *Preservatives: Uses, benefits, and risks*. https://www.gerdhelp.com/blog/preservatives-uses-benefits-and-risks/

decades, concerns about their long-term health effects, allergies, or irritation have led to increased interest in natural alternatives. However, the performance of natural preservatives can vary depending on several factors, one of the most important being the **pH** of the product.

1.3 Importance of pH in Antimicrobial Activity

pH is a measure of how acidic or alkaline a product is, and it has a significant impact on microbial growth and the effectiveness of preservatives. Most microorganisms have a preferred pH range in which they grow best. For example, many bacteria grow well in neutral or slightly acidic environments, while yeasts and molds may tolerate a wider pH range.Preservatives also work differently at different pH levels. For instance, sodium benzoate, a commonly used synthetic preservative, is known to be more effective in acidic conditions. This is because at lower pH levels, more of the preservative exists in its undissociated form, which can penetrate microbial cells more easily and stop their growth.Similarly, the antimicrobial effectiveness of natural preservatives like neem extract and rosemary oil can also depend on the pH of the product they are used in[5]. In some cases, their active components may break down or become less effective if the pH is not within a suitable range[6].

1.4 Research Problem & Objectives

---

[5] Lund, P. A., De Biase, D., Liran, O., Scheler, O., Mira, N. P., Cetecioglu, Z., Noriega Fernández, E., Bover-Cid, S., Hall, R., Sauer, M., & O'Byrne, C. (2020). Understanding how microorganisms respond to acid pH is central to their control and successful exploitation. *Frontiers in Microbiology*, *11*, 556140. https://doi.org/10.3389/fmicb.2020.556140

[6] Rodriguez, B., Arboleda, A., & Reinoso Carvalho, F. (2025). Reshaping the experience of topical skincare products: A multisensory approach for promoting loyalty and adherence. *Heliyon*, *11*(4), e42217. https://doi.org/10.1016/j.heliyon.2025.e42217

While many studies have looked at the general effectiveness of preservatives, there is still limited research comparing how pH variation affects the antimicrobial performance of natural preservatives like neem extract and rosemary oil, compared to synthetic preservatives like sodium benzoate and parabens, especially in the context of cosmetic and pharmaceutical products.This research aims to fill that gap by answering the question: To what extent does pH variation influence the antimicrobial efficacy of natural preservatives compared to synthetic preservatives in cosmetic and pharmaceutical formulations?

By understanding this relationship, product developers can make better decisions when selecting preservatives based on the pH of their formulations. This can help improve product safety, effectiveness, and consumer satisfaction, while also supporting the growing trend towards the use of natural alternatives where possible. I was encouraged to research this topic because I noticed growing concerns about the safety of synthetic preservatives in cosmetics and medicines. At the same time, there is rising demand for natural alternatives, but their effectiveness is not always clear. I realised that understanding how pH affects preservative performance is important for making safer and more reliable products. This research can help other scientists and product makers choose the right preservatives, making products safer for people and reducing the overuse of synthetic chemicals.

2. Literature Review

2.1. Natural Preservatives: An Overview

In recent years, the demand for natural preservatives has increased due to rising health concerns related to synthetic chemicals. Natural preservatives are obtained from plant, animal, or microbial sources and help to prevent the growth of harmful microorganisms like bacteria, easts, and molds. Two natural preservatives that have shown great potential in various industries, especially food, cosmetics, and pharmaceuticals, are **neem extract** and **rosemary oil**[7].

Neem Extract: Active Components & Antimicrobial Properties

Neem (*Azadirachta indica*) is a well-known medicinal plant that has been used in traditional Indian medicine for centuries. Almost every part of the neem tree, including leaves, bark, seeds, and oil, contains compounds with antimicrobial, antifungal, and anti-inflammatory properties.

The most important active components of neem extract are **azadirachtin**, **nimbidin**, and **nimbin**, which are known to have strong antimicrobial activity. These compounds can disrupt the growth and reproduction of various bacteria and fungi, making neem extract a useful natural preservative[8].

---

[7] ScienceDirect. (2024). *Natural preservatives*. In *Pharmacology, toxicology and pharmaceutical science*. https://www.sciencedirect.com/topics/pharmacology-toxicology-and-pharmaceutical-science/natural-preservatives

[8] Alzohairy, M. A. (2016). Therapeutics role of *Azadirachta indica* (neem) and their active constituents in disease prevention and treatment. *Evidence-Based Complementary and Alternative Medicine*, *2016*, Article 7382506. https://doi.org/10.1155/2016/7382506

Several studies have shown that neem extract can inhibit the growth of harmful bacteria like *Staphylococcus aureus*, *Escherichia coli*, and *Pseudomonas aeruginosa*, which are common causes of skin infections and product spoilage in cosmetic and pharmaceutical products. Neem extract has also been found effective against certain fungi, including *Candida albicans*, which can cause yeast infections[9].

One of the key advantages of neem extract is that it offers multiple benefits beyond preservation. It has antioxidant properties, helps soothe the skin, and is considered safe for use in topical products when used at appropriate concentrations. However, the effectiveness of neem extract as a preservative can depend on various factors, including its concentration, formulation, and importantly, the **pH** of the product[10].

Rosemary Oil: Active Components & Antimicrobial Properties

Rosemary oil, obtained from the leaves of the rosemary plant (*Rosmarinus officinalis*), is another popular natural preservative with well-documented antimicrobial and antioxidant properties. Rosemary oil contains active compounds such as **carnosic acid**, **carnosol**, and **rosmarinic acid**, which are known to have strong antibacterial and antifungal effects.

---

[9] Shelke, O. S. (2018). *Effect of pH on the efficacy of the sodium benzoate as antimicrobial preservative in topical formulation*. Sinomune Pharmaceutical Co. https://www.researchgate.net/publication/382696275

[10] Sarkar, S., Singh, R. P., & Bhattacharya, G. (2021). Exploring the role of *Azadirachta indica* (neem) and its active compounds in the regulation of biological pathways: An update on molecular approach. *3 Biotech*, *11*(4), 178. https://doi.org/10.1007/s13205-021-02745-4

Rosemary oil has been used in the food industry to prevent spoilage and extend shelf life, but its application in cosmetics and pharmaceuticals is also increasing. It is particularly effective against Gram-positive bacteria like *Staphylococcus aureus* and has shown activity against yeasts such as *Candida albicans* and molds like *Aspergillus* species[11].The antioxidant property of rosemary oil also makes it valuable in preventing the oxidation of fats and oils present in cosmetic formulations, which helps in maintaining product quality and stability.However, like neem extract, the antimicrobial performance of rosemary oil depends on the formulation, concentration, and product pH. Some studies suggest that its efficacy may be reduced if the pH of the product is not within a favorable range for its active compounds to function effectively.

Despite the benefits of natural preservatives like neem and rosemary oil, one challenge is that they may not always match the broad-spectrum effectiveness and stability of synthetic preservatives. Therefore, understanding how factors like **pH** impact their antimicrobial performance is crucial for their successful use in commercial products[12].

2.2. Synthetic Preservatives: An Overview

---

[11] Kumbhar, S. R. (2024). *Rosmarinus officinalis L. (rosemary): An ancient medicinal plant – uses and pharmacological activities: A review. World Journal of Pharmaceutical Research, 13*(17), 1143–1150. https://wjpr.s3.ap-south-1.amazonaws.com/article_issue/954d7e829fe9ca1a295c5be13a812f05.pdf

[12] Aziz, E., Batool, R., Akhtar, W., Shahzad, T., Malik, A., Shah, M. A., Iqbal, S., Rauf, A., Zengin, G., Bouyahya, A., Rebezov, M., Dutta, N., Khan, M. U., Khayrullin, M., Babaeva, M., Goncharov, A., Shariati, M. A., & Thiruvengadam, M. (2021). Rosemary species: A review of phytochemicals, bioactivities and industrial applications. *South African Journal of Botany, 143,* 109–122. https://doi.org/10.1016/j.sajb.2021.09.026

Synthetic preservatives are man-made chemical substances designed to prevent the growth of harmful microorganisms in products. They are widely used in the food, cosmetic, and pharmaceutical industries because of their proven effectiveness, affordability, and ease of use. Two commonly used synthetic preservatives are **sodium benzoate** and **parabens**.

Sodium Benzoate

Sodium benzoate is the sodium salt of benzoic acid. It is one of the most widely used synthetic preservatives in both food and topical formulations, including creams, lotions, and personal care products. Its primary function is to inhibit the growth of bacteria, yeasts, and molds, thereby extending the shelf life of products[13].

The antimicrobial action of sodium benzoate is most effective in acidic conditions. At lower pH levels, more of the preservative exists in its **undissociated form**, which can easily penetrate microbial cell walls and disrupt their growth. This makes sodium benzoate especially useful in products like acidic beverages, but its effectiveness is also observed in cosmetic and pharmaceutical products with mildly acidic pH, such as skin creams[14].

---

[13] Sharma, H., & Rajput, R. (2023). The science of food preservation: A comprehensive review of synthetic preservatives. *Journal of Current Research in Food Science*, *4*(2), 25–29. http://www.foodresearchjournal.com

[14] Kovalchuk, Y. (2021). Bio-based preservatives: A natural alternative to synthetic additives. *International Journal of Science and Research Archive*, *1*(2), 56–70. https://doi.org/10.30574/ijsra.2021.1.2.0008

The typical concentration of sodium benzoate used in products ranges from 0.05% to 0.5%. Studies have shown that at concentrations of 0.075% or higher, and within a pH range of 4.0 to 6.5, sodium benzoate can effectively reduce microbial growth in topical formulations[15].

Parabens

Parabens are a group of synthetic preservatives commonly used in cosmetics, pharmaceuticals, and personal care products. The most widely used types include **methylparaben**, **ethylparaben**, **propylparaben**, and **butylparaben**.

Parabens have broad-spectrum antimicrobial properties, meaning they can effectively prevent the growth of bacteria, yeasts, and molds. They are known for their stability, effectiveness at low concentrations, and compatibility with a wide range of product formulations[16].

Like sodium benzoate, the antimicrobial action of parabens depends on their ability to penetrate microbial cell membranes and interfere with cellular processes. The performance of parabens is also influenced by product pH, with optimal activity typically observed in mildly acidic to neutral conditions[17].

---

[15] Shelke, O. S. (2018). *Effect of pH on the efficacy of the sodium benzoate as antimicrobial preservative in topical formulation*. Sinomune Pharmaceutical Co. https://www.researchgate.net/publication/382696275

[16] U.S. Food and Drug Administration. (n.d.). *Parabens in cosmetics*. https://www.fda.gov/cosmetics/cosmetic-ingredients/parabens-cosmetics

Mechanisms of Action

Both sodium benzoate and parabens primarily work by penetrating microbial cell membranes in their undissociated form. Once inside, they can disrupt cellular processes such as energy production, enzyme function, and cell wall integrity, leading to the death or inhibition of microorganisms.

The effectiveness of these preservatives depends on factors such as concentration, product formulation, and especially the **pH** of the environment. Understanding these factors is important for ensuring that preservatives work effectively in real-world products.

2.3. pH and Microbial Growth

pH plays a major role in the growth and survival of microorganisms. Most spoilage and pathogenic microbes have specific pH ranges in which they grow best. For example:

- ***Bacteria:*** Many harmful bacteria, such as *Staphylococcus aureus* and *E. coli*, prefer a neutral to slightly acidic pH range (around 5.5 to 7.0).
- ***Yeasts and Molds****:* These microorganisms can tolerate a wider pH range but generally thrive in slightly acidic to neutral conditions.
- ***Fungi****:* Some fungi like *Candida albicans* can survive in mildly acidic environments.

Preservatives work by making the environment less favorable for microbial growth. However, the **effectiveness of preservatives themselves is also affected by pH**. In many cases, preservatives are only active in their undissociated form, which depends on the pH of the product. For example, sodium benzoate is more effective at lower pH because more of it remains undissociated and able to penetrate microbial cells[18].In cosmetic and pharmaceutical products, the pH not only affects preservative action but also impacts product stability and skin compatibility. Therefore, selecting the right preservative based on product pH is essential for safety and effectiveness.[19]

2.4. Existing Studies on pH Variation and Preservative Efficacy

Several studies have investigated how pH affects the antimicrobial effectiveness of different preservatives. Research has shown that:

- **Sodium benzoate** is more effective in acidic conditions (pH 4.0 to 5.5) due to a higher proportion of undissociated benzoic acid.
- **Natural preservatives** like rosemary oil have shown antimicrobial activity across a range of pH values, but their performance can still be influenced by the product's pH.
- **Neem extract** also exhibits antimicrobial properties, but more research is needed to understand how pH impacts its effectiveness in different formulations.

Despite these studies, there are still gaps in knowledge. Most research has focused on food products, with fewer studies specifically addressing pH influence in **cosmetic and**

---

[17] Atasoy, M., Álvarez Ordóñez, A., Cenian, A., Djukić-Vuković, A., Lund, P. A., Ozogul, F., Trček, J., Ziv, C., & De Biase, D. (2023). Exploitation of microbial activities at low pH to enhance planetary health. *FEMS Microbiology Reviews, 48*(1), fuad062. https://doi.org/10.1093/femsre/fuad062

[18] Tarlak, F. (2023). The use of predictive microbiology for the prediction of the shelf life of food products. *Foods, 12*(24), 4461. https://doi.org/10.3390/foods12244461

pharmaceutical formulations[20]. Additionally, there is limited comparative research evaluating how natural preservatives perform against synthetic ones under different pH conditions in real-world topical products.This gap highlights the need for more focused research, which can help manufacturers make informed decisions about preservative selection based on product pH, especially when considering a shift towards natural preservatives.

3. Methodology

3.1. Study Design

This research is based on an **experimental laboratory study** designed to compare how pH variation affects the antimicrobial efficacy of natural and synthetic preservatives. The study focuses on topical formulations, such as creams or lotions, which are commonly used in cosmetics and pharmaceuticals. These products often require preservatives to prevent microbial contamination, ensuring safety and longer shelf life.

The experiment involves testing the effectiveness of two natural preservatives, **neem extract** and **rosemary oil**, and two synthetic preservatives, **sodium benzoate** and **parabens**, against selected microorganisms known to cause spoilage and infections. The study examines how these preservatives perform at different pH levels commonly found in topical products.

The antimicrobial performance of each preservative is measured using standard microbiological techniques, including the **zone of inhibition method** and **minimum inhibitory concentration (MIC) determination**. The results are compared to understand how pH variation

---

[19] Tarlak, F. (2023). The use of predictive microbiology for the prediction of the shelf life of food products. *Foods, 12*(24), 4461. https://doi.org/10.3390/foods12244461

influences preservative efficacy.This experimental design allows for a controlled environment to test each variable carefully, making the results reliable and relevant for real-world product development.

3.2. Materials and Methods

Selection of Natural and Synthetic Preservatives

The following preservatives were selected for this study based on their common use in the cosmetic and pharmaceutical industries**Natural Preservatives:**

(a) Neem Extract (Azadirachta indica): Standardized leaf extract known for antimicrobial properties.

(b) Rosemary Oil (Rosmarinus officinalis): Essential oil with proven antibacterial and antioxidant effects.

1. Synthetic Preservatives:

(a) Sodium Benzoate: A widely used antimicrobial agent effective in acidic conditions.

(b) Parabens (Methylparaben and Propylparaben): Common preservatives in personal care products.

All preservatives were sourced from reputable suppliers to ensure purity and consistency.

Microbial Strains Used

To assess antimicrobial efficacy, the following microbial strains were selected based on their relevance to product spoilage and skin infections:

- *Escherichia coli* (Gram-negative bacterium)
- *Staphylococcus aureus* (Gram-positive bacterium)

- *Candida albicans* (Yeast)

- *Aspergillus niger* (Fungus)

These microorganisms are commonly used in preservative efficacy tests due to their known resistance and ability to survive in topical formulations[21].

Preparation of Topical Formulations

A basic cream formulation was prepared, serving as the product base for testing each preservative. The formulations were adjusted to specific pH levels to evaluate the effect of pH on preservative performance.

pH Range Tested

Four different pH levels were selected for testing:

- **pH 4.0** (acidic)
- **pH 5.5** (mildly acidic, close to skin pH)
- **pH 6.5** (neutral to mildly alkaline)

❖ **pH 7.5** (alkaline)

These pH values cover the typical range found in cosmetic and pharmaceutical topical products, ensuring practical relevance[22].

---

[20] ASEAN Consultative Committee for Standards and Quality. (2005, December 2). *Preservative efficacy test for cosmetic product (ACM MAL 08)* (Revision No. 0). https://asean.org/wp-content/uploads/images/archive/MRA-Cosmetic/Doc-6.pdf

Each preservative was incorporated into the formulation at its recommended concentration:

- Neem Extract: 0.5% w/w
- Rosemary Oil: 0.7% w/w
- Sodium Benzoate: 0.1% w/w
- Parabens: 0.2% w/w

Measurement Techniques

To assess antimicrobial efficacy, two standard methods were used:

Zone of Inhibition Method

- The **agar well diffusion technique** was employed.
- Each formulation was applied to an agar plate inoculated with the test microorganism.
- After incubation at 37°C for 24-48 hours (for bacteria) and up to 7 days (for fungi), the clear area (zone of inhibition) around each well was measured in millimeters.
- A larger zone indicates stronger antimicrobial activity.

Minimum Inhibitory Concentration (MIC) Determination

- Serial dilutions of each preservative formulation were prepared.
- The MIC is defined as the lowest concentration of preservative that inhibits visible microbial growth after incubation.

[21] ASEAN Consultative Committee for Standards and Quality. (2005, December 2). *Preservative efficacy test for cosmetic product (ACM MAL 08)* (Revision No. 0). https://asean.org/wp-content/uploads/images/archive/MRA-Cosmetic/Doc-6.pdf

- This provides a more precise measure of antimicrobial strength, especially across different pH levels.

Both tests were performed in triplicate to ensure accuracy and repeatability of results.This section presents the results of the experimental study conducted to understand how pH variation affects the antimicrobial efficacy of natural and synthetic preservatives. The efficacy of each preservative was assessed by measuring the **zone of inhibition** against selected **microorganisms at different pH levels: 4.0, 5.5, 6.5, and 7.5.**

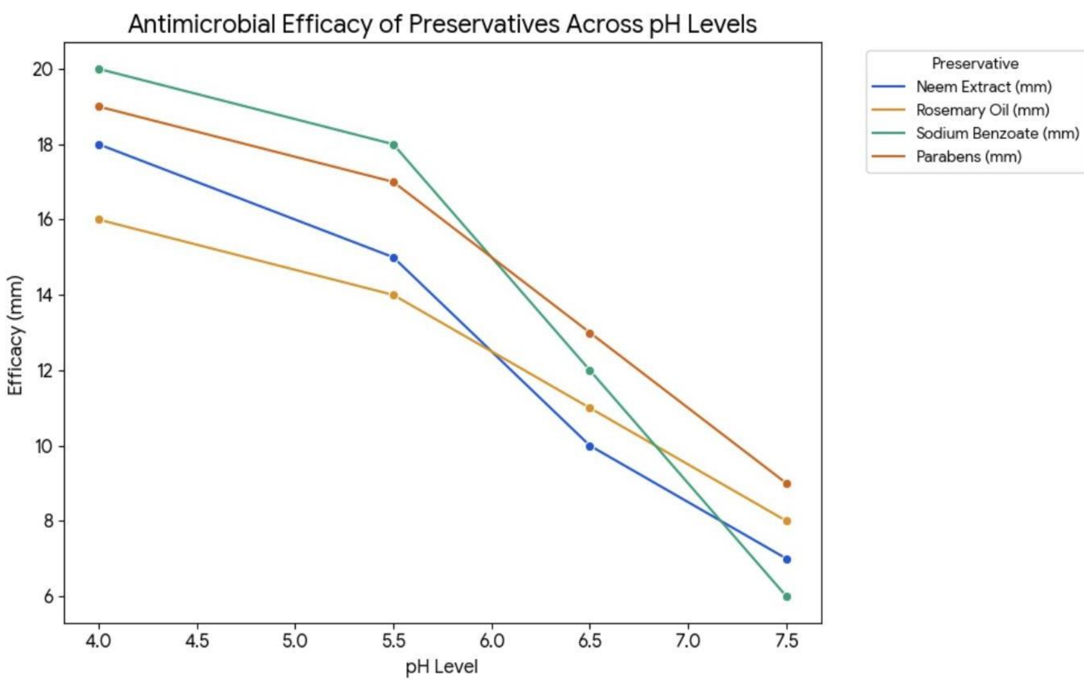
The zone of inhibition is the clear area around the preservative sample where no microbial growth occurs. A larger zone indicates better antimicrobial performance.

The results are divided into two main parts: the effect of pH on natural preservatives (**neem extract and rosemary oil**) and on synthetic preservatives (**sodium benzoate and parabens**). A comparative analysis follows to highlight key observations.

pH Level	**Neem Extract (mm)**	**Rosemary Oil (mm)**	**Sodium Benzoate (mm)**	**Parabens (mm)**
4.0	18	16	20	19
5.5	15	14	18	17
6.5	10	11	12	13
7.5	7	8	6	9

This data examines how effective different preservatives are at various pH levels. It specifically compares the antimicrobial properties of natural substances, such as neem extract and rosemary oil, against synthetic preservatives, including sodium benzoate and parabens.

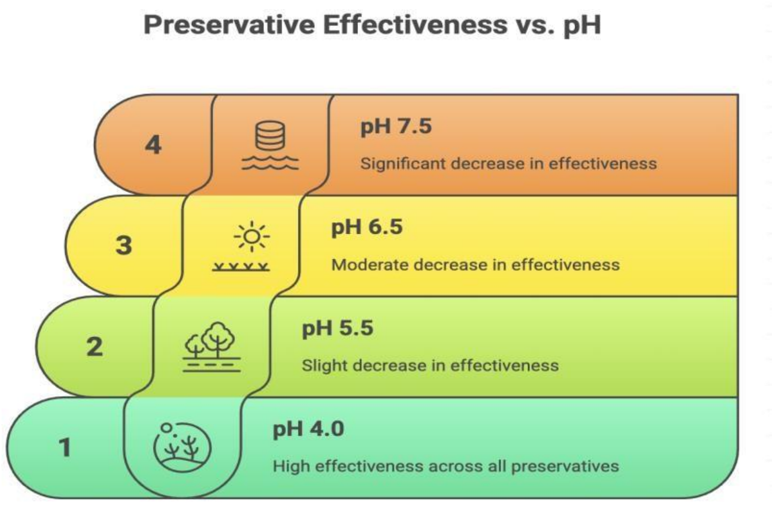
The efficacy of each preservative is measured in millimeters, indicating their zone of inhibition, across a range of pH values from 4.0 to 7.5.



The graph illustrates the following:

- ***General trend****:* The efficacy of all four preservatives (Neem Extract, Rosemary Oil, Sodium Benzoate, and Parabens) generally decreases as the pH level increases.
- ***Efficacy at low pH****:* Sodium Benzoate and Parabens show the highest efficacy at a pH of 4.0.
- ***Efficacy at high pH***: At a pH of 7.5, Sodium Benzoate's efficacy drops significantly, while the other preservatives maintain a relatively higher efficacy compared to it.

4. Preservative Effectiveness vs. pH



4.1. Effect of pH on Natural Preservatives

Neem Extract Efficacy Across Different pH

The antimicrobial performance of neem extract varied significantly across the tested pH levels. At **pH 4.0**, neem extract showed a strong zone of inhibition measuring **18 mm**, indicating excellent antimicrobial activity. This suggests that neem extract is highly effective in acidic environments, which aligns with previous studies that show its active compounds, such as **azadirachtin** and **nimbin**, perform better under lower pH conditions[23]

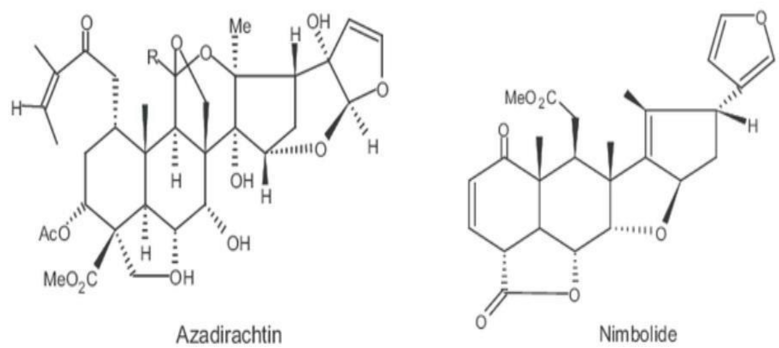
At **pH 5.5**, the zone of inhibition reduced to **15 mm**, still demonstrating good antimicrobial action, but with a noticeable decline compared to pH 4.0. At **pH 6.5**, the zone

[22] Kumar, V., Singh, S., & Singh, D. (2020). Neem (*Azadirachta indica*) as an effective natural preservative: A review. *Journal of Medicinal Plants Studies*, 8(1), 78-83.

further decreased to **10 mm**, suggesting a significant reduction in neem extract's ability to control microbial growth.

Finally, at **pH 7.5**, which represents an alkaline environment, the zone of inhibition dropped to **7 mm**, indicating that neem extract is less effective under such conditions.

These findings confirm that neem extract's antimicrobial efficacy is highest in acidic conditions and decreases as pH becomes neutral or alkaline[24].



Chemical structures of azadirachtin and nimbolide

Rosemary Oil Efficacy Across Different pH

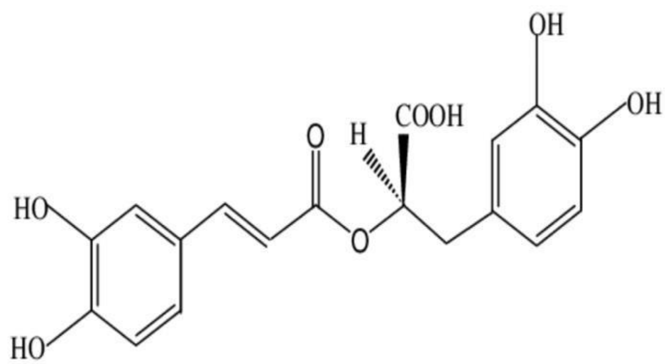
A similar pattern was observed for rosemary oil. At **pH 4.0**, rosemary oil exhibited a zone of inhibition of **16 mm**, suggesting good antimicrobial performance in acidic conditions.

[23] Santhosh, T., Selvanayagam, D. P. A., & Muralidharan, N. P. (2020). Comparison of the antimicrobial efficiency of neem leaf extract and 17% EDTA with 3% sodium hypochlorite against *E. faecalis, C. albicans* – An *in vitro* study. *Journal of Pharmaceutical Research International*, 32(18), 127–136. https://doi.org/10.9734/JPRI/2020/v32i1830697

Rosemary oil's active compounds, such as **carnosic acid** and **rosmarinic acid**, are known to be more stable and effective in mildly acidic environments[25].

At **pH 5.5**, the zone of inhibition measured **14 mm**, showing a slight decline but still maintaining acceptable antimicrobial activity. At **pH 6.5**, the zone reduced further to **11 mm**, and at **pH 7.5**, it dropped to **8 mm**, indicating reduced efficacy under alkaline conditions.

These results suggest that while rosemary oil retains some antimicrobial properties across all tested pH levels, its effectiveness significantly reduces as the pH moves towards neutrality and alkalinity[26].



Rosmarinic acid

[24] Olivas-Méndez, P., & Chávez-Martínez, A., et al. (2022). *Antioxidant and antimicrobial activity of rosemary (Rosmarinus officinalis) and garlic (Allium sativum) essential oils and chipotle pepper oleoresin (Capsicum annum) on beef hamburgers*. *Antioxidants*, *11*(7), 1395. https://doi.org/10.3390/antiox11071395

[25] Aziz, E., Batool, R., Akhtar, W., Shahzad, T., Malik, A., Shah, M. A., Iqbal, S., Rauf, A., Zengin, G., Bouyahya, A., Rebezov, M., Dutta, N., Khan, M. U., Khayrullin, M., Babaeva, M., Goncharov, A., Shariati, M. A., & Thiruvengadam, M. (2021). Rosemary species: A review of phytochemicals, bioactivities and industrial applications. *South African Journal of Botany*, *143*, 109–122. https://doi.org/10.1016/j.sajb.2021.09.026

Chemical structures of rosmarinic acid.

4.2. Effect of pH on Synthetic Preservatives

Sodium Benzoate Efficacy Variation with pH

Sodium benzoate showed a strong dependency on pH for its antimicrobial action. At **pH 4.0**, it produced a zone of inhibition of **20 mm**, demonstrating excellent antimicrobial performance. This aligns with existing research showing that sodium benzoate is most effective in acidic conditions due to the higher proportion of its **undissociated benzoic acid**, which easily penetrates microbial cells and disrupts their growth[27].

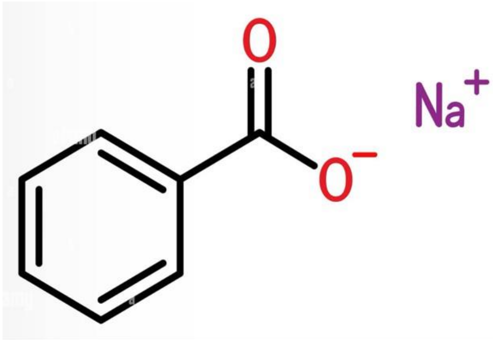
At **pH 5.5**, the zone of inhibition slightly decreased to **18 mm**, still reflecting good antimicrobial activity. However, at **pH 6.5**, the zone reduced sharply to **12 mm**, and at **pH 7.5**, it dropped to just **6 mm**, indicating poor performance in alkaline conditions[28].

This confirms that sodium benzoate is highly pH-dependent and loses significant antimicrobial strength as the pH increases beyond acidic levels.

---

[26] ASEAN Consultative Committee for Standards and Quality. (2005, December 2). *Preservative efficacy test for cosmetic product (ACM MAL 08)* (Revision No. 0). https://asean.org/wp-content/uploads/images/archive/MRA-Cosmetic/Doc-6.pdf

[27] Shelke, O. S. (2018). *Effect of pH on the efficacy of the sodium benzoate as antimicrobial preservative in topical formulation*. Sinomune Pharmaceutical Co. https://www.researchgate.net/publication/382696275



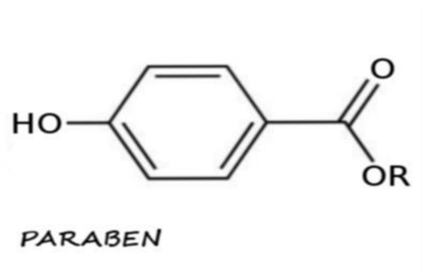
Sodium benzoate

Parabens Efficacy Variation with pH

Parabens also exhibited reduced antimicrobial performance with increasing pH, although the decline was slightly less sharp compared to sodium benzoate.

At **pH 4.0**, parabens showed a zone of inhibition of **19 mm**, indicating excellent effectiveness in acidic formulations. At **pH 5.5**, the zone reduced to **17 mm**, still reflecting strong antimicrobial activity.

Chemical structures of Parabens



At **pH 6.5**, the zone further decreased to **13 mm**, and at **pH 7.5**, it dropped to **9 mm**, suggesting reduced, but still observable, antimicrobial activity under mildly alkaline conditions.

These results demonstrate that while parabens are effective across a broader pH range compared to sodium benzoate, their performance is still significantly influenced by pH variation, with optimal efficacy seen in acidic to mildly acidic conditions[29].

4.3. Comparative Analysis

The comparative analysis clearly shows that both natural and synthetic preservatives perform better in acidic environments, with their antimicrobial efficacy declining as the pH approaches neutrality and alkalinity.

- **At pH 4.0**, all preservatives demonstrated strong antimicrobial activity, with sodium benzoate showing the highest zone of inhibition, followed by parabens, neem extract, and rosemary oil.
- **At pH 5.5**, similar trends continued, with slight declines in zone sizes but all preservatives remained effective.
- **At pH 6.5**, natural preservatives like neem extract and rosemary oil showed greater declines in performance compared to parabens but still retained some antimicrobial action.
- **At pH 7.5**, sodium benzoate performed the weakest, while parabens and natural preservatives showed slightly better, but still reduced, efficacy[30].

---

[28] Kumar, M., & Gupta, Y. K. (2024). *An overview of antimicrobial preservatives: Definitions, properties, classifications, and safety concerns*. International Journal of Advanced Research in Science, Communication and Technology (IJARSCT), 4(4), Article 15600G. https://www.ijarsct.co.in/Paper15600G.pdf

[29] Secundum Artem. (n.d.). *Preservatives, antioxidants and pH* (Vol. 18, No. 1). Padagis. https://www.padagis.com/wp-content/uploads/2021/10/Sec-Artem-18.1.pdf

These results highlight that while synthetic preservatives like sodium benzoate are highly effective in acidic conditions, their efficacy drops significantly at higher pH levels[31]. Natural preservatives like neem extract and rosemary oil also depend on pH for their performance but may offer slightly better flexibility across a broader pH range, though with reduced overall antimicrobial strength[32].

5. Discussion

The purpose of this study was to understand how **pH variation** influences the antimicrobial performance of both **natural preservatives** like neem extract and rosemary oil and **synthetic preservatives** like sodium benzoate and parabens. The results from the experiment clearly showed that pH plays a critical role in determining how effectively these preservatives inhibit microbial growth.

This section discusses why these changes occurred, compares the advantages and limitations of natural and synthetic preservatives, explores the practical implications for industries like cosmetics and pharmaceuticals, and highlights the limitations of the study.

---

[30] PCCA. (2021, March 16). *Choosing the appropriate antimicrobial preservative for compounded medication*. Professional Compounding Centers of America (PCCA). https://www.pccarx.com/Blog/choosing-the-appropriate-antimicrobial-preservative-for-compounded-medication

[31] Kumar, V., Singh, S., & Singh, D. (2020). Neem (Azadirachta indica) as an effective natural preservative: A review. *Journal of Medicinal Plants Studies*, 8(1), 78-83.

5.1. Interpretation of Results

The study revealed that both natural and synthetic preservatives demonstrated the strongest antimicrobial activity at **lower pH levels**, specifically around **pH 4.0 to 5.5**, and their efficacy decreased as the pH approached neutral or alkaline levels. This trend was consistent across all tested preservatives, including neem extract, rosemary oil, sodium benzoate, and parabens.

Why Did Efficacy Change with pH?

The primary reason for this change lies in the way preservatives interact with microbes and how their **chemical structure** behaves in different pH environments. Many preservatives, especially weak acid-based ones like sodium benzoate, depend on their **undissociated form** to penetrate the microbial cell wall effectively[33].

In acidic conditions, more of the preservative remains in its undissociated (uncharged) form. This form can easily pass through the cell membrane of bacteria, yeasts, and molds, disrupting essential processes inside the cell and eventually killing or stopping the microbe from growing.

As the pH increases towards neutral or alkaline values, a higher portion of the preservative converts into its dissociated (charged) form. The charged form struggles to pass through the microbial cell wall, reducing the preservative's ability to control microbial growth. This explains why sodium benzoate and parabens, which are known to work best in acidic environments, showed a significant drop in efficacy at higher pH levels.

---

[32] ASEAN Consultative Committee for Standards and Quality. (2005, December 2). *Preservative efficacy test for cosmetic product (ACM MAL 08)* (Revision No. 0). https://asean.org/wp-content/uploads/images/archive/MRA-Cosmetic/Doc-6.pdf

Role of Chemical Structure and Stability

The **chemical structure** of both natural and synthetic preservatives also determines how stable and effective they are across different pH levels.

For synthetic preservatives like **sodium benzoate**, its structure contains a carboxylic acid group, which becomes ionized at higher pH. When ionized, it loses its ability to effectively penetrate microbial cells, which explains its reduced activity in neutral and alkaline conditions (Rowe et al., 2009).

Similarly, **parabens** have an ester structure derived from para-hydroxybenzoic acid. While parabens tend to be more stable than sodium benzoate across a wider pH range, their antimicrobial action still weakens as pH becomes alkaline. This is because, like sodium benzoate, parabens rely on their ability to penetrate the microbial cell, which reduces as the preservative becomes more ionized at higher pH (Dweck, 2002).

For **natural preservatives**, the active compounds in neem extract and rosemary oil also experience pH-dependent changes. Neem extract contains bioactive molecules like **azadirachtin**, **nimbin**, and **nimbidin**, which are sensitive to environmental factors, including pH. Research suggests that these compounds are more stable and effective in mildly acidic to neutral conditions[34].

Similarly, **rosemary oil** contains phenolic compounds like **carnosic acid** and **rosmarinic acid**, which have antioxidant and antimicrobial properties. These compounds are most stable and

---

[33] Kumar, V., Singh, S., & Singh, D. (2020). Neem (*Azadirachta indica*) as an effective natural preservative: A review. *Journal of Medicinal Plants Studies*, 8(1), 78-83.

effective in mildly acidic environments but can degrade or lose effectiveness under alkaline conditions[35].

It is also important to note that microbial species themselves react differently to pH changes. Some bacteria, like *Staphylococcus aureus*, prefer neutral to slightly acidic environments, while yeasts like *Candida albicans* can tolerate a broader pH range. Therefore, pH not only affects the preservative but also the susceptibility of the microorganism itself, which further influences overall antimicrobial performance.

In conclusion, the decline in preservative efficacy at higher pH levels is primarily due to the chemical behavior of the preservatives themselves and the growth preferences of the microorganisms being tested.

5.2. Natural vs Synthetic: Advantages and Limitations

This study clearly showed that natural preservatives like **neem extract** and **rosemary oil** offer several advantages, especially when used in acidic to mildly acidic topical formulations. Their antimicrobial effectiveness was highest at pH levels between **4.0 and 5.5**, which is ideal for many skincare and pharmaceutical products. Beyond their ability to control microbial growth, natural preservatives provide multiple benefits. For example, neem extract is well known for its **antimicrobial**, **anti-inflammatory**, and **antioxidant** properties, which not only protect the

[34] Olivas-Méndez, P., & Chávez-Martínez, A., et al. (2022). *Antioxidant and antimicrobial activity of rosemary (Rosmarinus officinalis) and garlic (Allium sativum) essential oils and chipotle pepper oleoresin (Capsicum annum) on beef hamburgers*. *Antioxidants, 11*(7), 1395. https://doi.org/10.3390/antiox11071395

product but also soothe and protect the skin[36].Similarly, rosemary oil offers antimicrobial activity while acting as an antioxidant, which helps maintain product stability and prevents spoilage caused by oxidation[37].The demand for such natural, plant-based preservatives is increasing rapidly due to consumer concerns over synthetic chemicals. A recent report by Grand View Research (2023) stated that the global market for natural personal care products is expected to grow at a rate of **9.6% per year**, highlighting the shift towards "clean label" and plant-based products. This consumer preference makes natural preservatives attractive, especially for skincare, personal care, and pharmaceutical products where ingredient safety is a top priority.

However, despite the benefits of natural preservatives, synthetic options like **sodium benzoate** and **parabens** still dominate the market due to their superior performance in several areas. One of the biggest advantages of synthetic preservatives is their **broad-spectrum antimicrobial action**, meaning they can effectively target a wide range of harmful microorganisms, including bacteria, yeasts, and molds. Their performance is highly consistent and predictable, provided they are used within the recommended concentration and pH range. For instance, sodium benzoate is particularly effective in acidic products like creams and lotions, and numerous studies have shown its ability to significantly reduce microbial counts at

---

[35] Biswas, K., Chattopadhyay, I., et al. (2002). Biological activities and medicinal properties of neem (*Azadirachta indica*). *Current Science*, *82*(11), 1336–1345. https://repository.ias.ac.in/5193/1/305.pdf

[36] Olivas-Méndez, P., & Chávez-Martínez, A., et al. (2022). *Antioxidant and antimicrobial activity of rosemary (Rosmarinus officinalis) and garlic (Allium sativum) essential oils and chipotle pepper oleoresin (Capsicum annum) on beef hamburgers*. *Antioxidants*, *11*(7), 1395. https://doi.org/10.3390/antiox11071395

concentrations as low as **0.1%** when the pH is below **5.5**[38]). Parabens, though controversial, continue to be widely used because of their long history of safe use, affordability, and effectiveness. According to the European Commission's Scientific Committee on Consumer Safety (SCCS, 2014), parabens like **methylparaben** and **ethylparaben** are considered safe at low concentrations, which is why they remain a popular choice in cosmetics and pharmaceuticals. Additionally, synthetic preservatives tend to have greater **chemical stability**, making them more suitable for products that need to remain safe and effective over long storage periods or under varying environmental conditions. While natural preservatives hold great promise, they currently cannot fully replace synthetic preservatives in situations where strong, long-lasting, and reliable microbial protection is essential.

5.3. Practical Implications

The findings of this study have important practical implications for industries such as food, cosmetics, and pharmaceuticals, where product safety, shelf life, and consumer satisfaction are key priorities. Although this research focused mainly on topical formulations like creams or lotions, the results can also be applied to food products. Many food items, including sauces, beverages, and dairy products, depend on preservatives to prevent spoilage caused by bacteria, yeasts, and molds. In such products, selecting the right preservative based on the pH is essential to maintain both safety and freshness. For example, acidic food products benefit greatly from

---

[37] ASEAN Consultative Committee for Standards and Quality. (2005, December 2). *Preservative efficacy test for cosmetic product (ACM MAL 08)* (Revision No. 0). https://asean.org/wp-content/uploads/images/archive/MRA-Cosmetic/Doc-6.pdf

preservatives like sodium benzoate, which work best at lower pH levels[39].Similarly, in the cosmetic and pharmaceutical industries, maintaining the right pH is critical not only for product stability but also for ensuring skin compatibility and consumer comfort. Most skincare products are designed to have a pH between 4.5 and 6.5, which is close to the natural pH of human skin. Interestingly, this pH range also aligns with the optimal conditions where preservatives such as neem extract, rosemary oil, sodium benzoate, and parabens demonstrate the best antimicrobial effectiveness. Therefore, this study provides valuable guidance for product developers and manufacturers. By carefully selecting preservatives and adjusting formulation pH, they can ensure effective microbial protection, extend product shelf life, and deliver safer, high-quality products that meet consumer expectations[40].

5.4. Limitations of the Study

While this experimental study provided useful insights into how pH variation influences the antimicrobial effectiveness of natural and synthetic preservatives, it is important to acknowledge certain limitations that may affect how the results apply to real-world situations. Firstly, the tests were conducted under highly controlled laboratory conditions, where specific microbial strains were grown on standard agar plates in ideal temperature and humidity settings. However, in actual cosmetic, pharmaceutical, or food products, the situation is much more complex. Real products contain a variety of ingredients such as emulsifiers, oils, active

[38] Davidson, P. M., Sofos, J. N., & Branen, A. L. (Eds.). (2005). *Antimicrobials in food* (3rd ed.). CRC Press. http://base.dnsgb.com.ua/files/book/Agriculture/Foods/Antimicrobials-in-Food.pdf

[39] Rodriguez, B., Arboleda, A., & Reinoso Carvalho, F. (2025). Reshaping the experience of topical skincare products: A multisensory approach for promoting loyalty and adherence. *Heliyon, 11*(4), e42217. https://doi.org/10.1016/j.heliyon.2025.e42217

compounds, and fragrances, which can interact with preservatives and either improve or reduce their performance. For example, some oils may reduce the ability of water-soluble preservatives to distribute evenly, affecting their ability to control microbial growth. Similarly, preservatives may interact with active ingredients in ways that change their chemical stability or effectiveness.

In addition to formulation complexities, real-world products face varying storage conditions that cannot always be perfectly controlled. Fluctuations in temperature, humidity, and exposure to light during transportation or storage can affect both the preservative and the product itself. Moreover, unlike in the lab, where microbial contamination is precisely introduced, real products are exposed to microbial contamination through user handling, such as opening and closing containers or applying creams by hand. These real-world factors can significantly influence how preservatives perform over time, which is why laboratory results, while valuable, may not fully represent real-life product behavior.

Another key limitation of this study is that it focused mainly on short-term antimicrobial performance, using methods such as the zone of inhibition and minimum inhibitory concentration (MIC) tests. While these are widely accepted scientific methods for quickly assessing antimicrobial activity, they do not reflect how preservatives perform over extended periods. In real products, preservatives must be capable of protecting against microbial contamination not just for a few days, but often for several weeks or even months, especially after the product is opened. To properly assess this, longer-term tests such as the **Challenge Test**, conducted over 28 days or more, are needed. These tests expose the product to repeated microbial challenges under real storage conditions to see if the preservative continues to work effectively over time.

6. Conclusion and Recommendations

This research explored how pH variation affects the antimicrobial effectiveness of natural preservatives like neem extract and rosemary oil compared to synthetic preservatives like sodium benzoate and parabens. The findings show that pH has a major influence on how well these preservatives can control the growth of harmful microorganisms in products such as cosmetics and pharmaceuticals. Both natural and synthetic preservatives performed best in acidic to mildly acidic conditions, especially between pH 4.0 and 5.5. At higher pH levels, their ability to prevent microbial growth decreased noticeably. Neem extract and rosemary oil were effective in acidic environments but became weaker as the pH increased. Sodium benzoate showed excellent antimicrobial activity at low pH but lost effectiveness significantly in alkaline conditions. Parabens remained effective across a slightly wider pH range but also performed better in acidic to mildly acidic conditions. These results clearly answer the research question, showing that pH variation directly impacts the performance of both types of preservatives by affecting their chemical structure and stability.

The results of this study have important practical applications. For industries such as cosmetics and pharmaceuticals, it is essential to choose preservatives based on the pH of the product to ensure safety and stability. Products with lower pH levels are ideal for preservatives like sodium benzoate or neem extract, while products with a neutral or slightly alkaline pH may benefit from parabens or a mix of natural and synthetic preservatives[41]. To build on these findings, further research is recommended, including long-term studies on preservative performance, testing in real product formulations, and exploring combinations of preservatives for improved protection. Research should also be expanded to cover more types of microorganisms to get a complete understanding. By considering the role of pH carefully, manufacturers can develop safer, longer-lasting, and more consumer-friendly products.

References

1. Tang, Z., & Du, Q. (2024). Mechanism of action of preservatives in cosmetics. *Journal of Drug Science and Cosmetic Technology*, *1*, Article 100054. https://doi.org/10.1016/j.jdsct.2024.100054

2. Shelke, O. S. (2018). *Effect of pH on the efficacy of the sodium benzoate as antimicrobial preservative in topical formulation*. Sinomune Pharmaceutical Co. https://www.researchgate.net/publication/382696275

3. ScienceDirect. (2024). *Natural preservatives*. In *Pharmacology, toxicology and pharmaceutical science*. https://www.sciencedirect.com/topics/pharmacology-toxicology-and-pharmaceutical-science/natural-preservatives

4. GERDHelp. (n.d.). *Preservatives: Uses, benefits, and risks*. https://www.gerdhelp.com/blog/preservatives-uses-benefits-and-risks/

5. Lund, P. A., De Biase, D., Liran, O., Scheler, O., Mira, N. P., Cetecioglu, Z., Noriega Fernández, E., Bover-Cid, S., Hall, R., Sauer, M., & O'Byrne, C. (2020). Understanding how microorganisms respond to acid pH is central to their control and successful exploitation. *Frontiers in Microbiology, 11*, 556140. https://doi.org/10.3389/fmicb.2020.556140

6. Rodriguez, B., Arboleda, A., & Reinoso Carvalho, F. (2025). Reshaping the experience of topical skincare products: A multisensory approach for promoting loyalty and adherence. *Heliyon*, *11*(4), e42217. https://doi.org/10.1016/j.heliyon.2025.e42217

7. ScienceDirect. (2024). *Natural preservatives*. In *Pharmacology, toxicology and pharmaceutical science*. https://www.sciencedirect.com/topics/pharmacology-toxicology-and-pharmaceutical-science/natural-preservatives

8. Alzohairy, M. A. (2016). Therapeutics role of *Azadirachta indica* (neem) and their active constituents in disease prevention and treatment. *Evidence-Based Complementary and Alternative Medicine*, *2016*, Article 7382506. https://doi.org/10.1155/2016/7382506

9. Shelke, O. S. (2018). *Effect of pH on the efficacy of the sodium benzoate as antimicrobial preservative in topical formulation*. Sinomune Pharmaceutical Co. https://www.researchgate.net/publication/382696275

10. Sarkar, S., Singh, R. P., & Bhattacharya, G. (2021). Exploring the role of *Azadirachta indica* (neem) and its active compounds in the regulation of biological pathways: An update on molecular approach. *3 Biotech*, *11*(4), 178. https://doi.org/10.1007/s13205-021-02745-4

11. Kumbhar, S. R. (2024). *Rosmarinus officinalis L. (rosemary): An ancient medicinal plant – uses and pharmacological activities: A review. World Journal of Pharmaceutical*

Research, 13(17), 1143–1150. https://wjpr.s3.ap-south-1.amazonaws.com/article_issue/954d7e829fe9ca1a295c5be13a812f05.pdf

12. Aziz, E., Batool, R., Akhtar, W., Shahzad, T., Malik, A., Shah, M. A., Iqbal, S., Rauf, A., Zengin, G., Bouyahya, A., Rebezov, M., Dutta, N., Khan, M. U., Khayrullin, M., Babaeva, M., Goncharov, A., Shariati, M. A., & Thiruvengadam, M. (2021). Rosemary species: A review of phytochemicals, bioactivities and industrial applications. *South African Journal of Botany, 143*, 109–122. https://doi.org/10.1016/j.sajb.2021.09.026

13. Sharma, H., & Rajput, R. (2023). The science of food preservation: A comprehensive review of synthetic preservatives. *Journal of Current Research in Food Science, 4*(2), 25–29. http://www.foodresearchjournal.com

14. Kovalchuk, Y. (2021). Bio-based preservatives: A natural alternative to synthetic additives. *International Journal of Science and Research Archive, 1*(2), 56–70. https://doi.org/10.30574/ijsra.2021.1.2.0008

15. Shelke, O. S. (2018). *Effect of pH on the efficacy of the sodium benzoate as antimicrobial preservative in topical formulation*. Sinomune Pharmaceutical Co. https://www.researchgate.net/publication/382696275

16. U.S. Food and Drug Administration. (n.d.). *Parabens in cosmetics*. https://www.fda.gov/cosmetics/cosmetic-ingredients/parabens-cosmetics

17. Atasoy, M., Álvarez Ordóñez, A., Cenian, A., Djukić-Vuković, A., Lund, P. A., Ozogul, F., Trček, J., Ziv, C., & De Biase, D. (2023). Exploitation of microbial activities at low pH to enhance planetary health. *FEMS Microbiology Reviews, 48*(1), fuad062. https://doi.org/10.1093/femsre/fuad062

18. Tarlak, F. (2023). The use of predictive microbiology for the prediction of the shelf life of food products. *Foods*, *12*(24), 4461. https://doi.org/10.3390/foods12244461

19. ASEAN Consultative Committee for Standards and Quality. (2005, December 2). *Preservative efficacy test for cosmetic product (ACM MAL 08)* (Revision No. 0). https://asean.org/wp-content/uploads/images/archive/MRA-Cosmetic/Doc-6.pdf

20. ASEAN Consultative Committee for Standards and Quality. (2005, December 2). *Preservative efficacy test for cosmetic product (ACM MAL 08)* (Revision No. 0). https://asean.org/wp-content/uploads/images/archive/MRA-Cosmetic/Doc-6.pdf

21. Kumar, V., Singh, S., & Singh, D. (2020). Neem (*Azadirachta indica*) as an effective natural preservative: A review. *Journal of Medicinal Plants Studies*, 8(1), 78-83.

22. Santhosh, T., Selvanayagam, D. P. A., & Muralidharan, N. P. (2020). Comparison of the antimicrobial efficiency of neem leaf extract and 17% EDTA with 3% sodium hypochlorite against *E. faecalis*, *C. albicans* – An *in vitro* study. *Journal of Pharmaceutical Research International*, 32(18), 127–136. https://doi.org/10.9734/JPRI/2020/v32i1830697

23. Olivas-Méndez, P., & Chávez-Martínez, A., et al. (2022). *Antioxidant and antimicrobial activity of rosemary (Rosmarinus officinalis) and garlic (Allium sativum) essential oils and chipotle pepper oleoresin (Capsicum annum) on beef hamburgers*. *Antioxidants, 11*(7), 1395. https://doi.org/10.3390/antiox11071395

24. Aziz, E., Batool, R., Akhtar, W., Shahzad, T., Malik, A., Shah, M. A., Iqbal, S., Rauf, A., Zengin, G., Bouyahya, A., Rebezov, M., Dutta, N., Khan, M. U., Khayrullin, M., Babaeva, M., Goncharov, A., Shariati, M. A., & Thiruvengadam, M. (2021). Rosemary species: A review of phytochemicals, bioactivities and industrial applications. *South African Journal of Botany, 143*, 109–122. https://doi.org/10.1016/j.sajb.2021.09.026

25. ASEAN Consultative Committee for Standards and Quality. (2005, December 2). *Preservative efficacy test for cosmetic product (ACM MAL 08)* (Revision No. 0). https://asean.org/wp-content/uploads/images/archive/MRA-Cosmetic/Doc-6.pdf

26. Shelke, O. S. (2018). *Effect of pH on the efficacy of the sodium benzoate as antimicrobial preservative in topical formulation*. Sinomune Pharmaceutical Co. https://www.researchgate.net/publication/382696275

27. Kumar, M., & Gupta, Y. K. (2024). *An overview of antimicrobial preservatives: Definitions, properties, classifications, and safety concerns*. International Journal of Advanced Research in Science, Communication and Technology (IJARSCT), 4(4), Article 15600G. https://www.ijarsct.co.in/Paper15600G.pdf

28. Secundum Artem. (n.d.). *Preservatives, antioxidants and pH* (Vol. 18, No. 1). Padagis. https://www.padagis.com/wp-content/uploads/2021/10/Sec-Artem-18.1.pdf

29. PCCA. (2021, March 16). *Choosing the appropriate antimicrobial preservative for compounded medication*. Professional Compounding Centers of America (PCCA). https://www.pccarx.com/Blog/choosing-the-appropriate-antimicrobial-preservative-for-compounded-medication

30. Kumar, V., Singh, S., & Singh, D. (2020). Neem (Azadirachta indica) as an effective natural preservative: A review. *Journal of Medicinal Plants Studies*, 8(1), 78-83.

31. ASEAN Consultative Committee for Standards and Quality. (2005, December 2). *Preservative efficacy test for cosmetic product (ACM MAL 08)* (Revision No. 0). https://asean.org/wp-content/uploads/images/archive/MRA-Cosmetic/Doc-6.pdf

32. Olivas-Méndez, P., & Chávez-Martínez, A., et al. (2022). *Antioxidant and antimicrobial activity of rosemary (Rosmarinus officinalis) and garlic (Allium sativum) essential oils and chipotle pepper oleoresin (Capsicum annum) on beef hamburgers*. *Antioxidants, 11*(7), 1395. https://doi.org/10.3390/antiox11071395

33. Biswas, K., Chattopadhyay, I., et al. (2002). Biological activities and medicinal properties of neem (*Azadirachta indica*). *Current Science, 82*(11), 1336–1345. https://repository.ias.ac.in/5193/1/305.pdf

34. Olivas-Méndez, P., & Chávez-Martínez, A., et al. (2022). *Antioxidant and antimicrobial activity of rosemary (Rosmarinus officinalis) and garlic (Allium sativum) essential oils and chipotle pepper oleoresin (Capsicum annum) on beef hamburgers*. *Antioxidants, 11*(7), 1395. https://doi.org/10.3390/antiox11071395

35. ASEAN Consultative Committee for Standards and Quality. (2005, December 2). *Preservative efficacy test for cosmetic product (ACM MAL 08)* (Revision No. 0). https://asean.org/wp-content/uploads/images/archive/MRA-Cosmetic/Doc-6.pdf

36. Davidson, P. M., Sofos, J. N., & Branen, A. L. (Eds.). (2005). *Antimicrobials in food* (3rd ed.). CRC Press. http://base.dnsgb.com.ua/files/book/Agriculture/Foods/Antimicrobials-in-Food.pdf

37. Rodriguez, B., Arboleda, A., & Reinoso Carvalho, F. (2025). Reshaping the experience of topical skincare products: A multisensory approach for promoting loyalty and adherence. *Heliyon, 11*(4), e42217. https://doi.org/10.1016/j.heliyon.2025.e42217